1) Express each of the following as a single fraction in its simplest form.

(a) $\frac{1}{a+b}+\frac{a}{4(a+b)}$

(b) $\frac{2}{x+1}+\frac{3}{(x+1)(x-2)}$

(c) $\frac{7}{x+3}-\frac{4}{(x+3)(x-5)}$

(d) $\frac{3}{x-2}-\frac{5}{(x-2)(x+4)}$

(e) $\frac{5}{3-a}+\frac{6}{(3-a)(3+a)}$

(f) $\frac{3}{a-4}+\frac{a}{a^2-16}$

(g) $\frac{p}{p^2-9}+\frac{2}{p+3}$

(h) $\frac{7}{5a-5b}+\frac{4}{2a-2b}$

(i) $\frac{c}{25-c^2}-\frac{1}{5-c}$

(j) $\frac{6}{x-3}+\frac{x}{x^2-3x}$

2) Simplify each of the following:

(a) $\frac{a^2}{a^2+3a}\times\frac{a+3}{a}$

(b) $\frac{3x-2}{4x}\times\frac{24}{18x-12}$

(c) $\frac{49-b^2}{b^2}\times\frac{b}{7-b}$

(d) $\frac{3u^2+7u+2}{21}\times\frac{7u}{u+2}$

(e) $\frac{2y-1}{y^2}\times\frac{y}{2y^2+5y-3}$

(f) $\frac{2p^2}{p^2+8p+7}\times\frac{p+7}{4p}$

(g) $\frac{2x-6}{4}\times\frac{x}{x^2-x-6}$

(h) $\frac{3}{y^2+y-2}\times\frac{3y+6}{y}$

(i) $\frac{3}{p^2-9}\div\frac{4}{p+3}$

(j) $\frac{25-n^2}{5}\div\frac{5-n}{2}$

(k) $\frac{a^2-16}{a^2}\div\frac{a+4}{5a}$

(l) $\frac{2y^2-y}{4y}\div\frac{2y-1}{10}$

3) Show that

$$\frac{3x}{x+2}-\frac{2x+1}{x-2}-1$$

can be written in the form

$$\frac{ax+b}{x^2-4}$$

where a and b are integers. **[Edexcel - June 2022]**

4) Show that

$$\frac{6x^3}{9x^2-144} \div \frac{2x^2}{3(x-4)}$$

Can be written in the form

$$\frac{1}{x(x-r)}$$

Where r is an integer. **[Edexcel - Nov 2021]**

5) Show that

$$\frac{7x-14}{x^2+4x-12} \div \frac{x-6}{x^3-36x}$$

Simplifies to ax where a is an integer. [Edexcel – Nov 2019]

6) Show that

$$6+\left[(x+5) \div \frac{x^2+3x-10}{x-1}\right]$$

Can be expressed in the form $\frac{ax-b}{cx-d}$, where a,b,c and d are integers.

7) **[Edexcel – Nov 2018]**

a) Write

$$\frac{4x^2-9}{6x+9} \times \frac{2x}{x^2-3x}$$

In the form

$$\frac{ax+b}{cx+d}$$

Where a,b,c and d are integers.

b) Express

$$\frac{3}{x+1}+\frac{1}{x-2}-\frac{4}{x}$$

As a single fraction in its simplest form.

8) **[Edexcel – June 2017]**

$$2-\frac{x+2}{x-3}-\frac{x-6}{x+3}$$

Can be written as a single fraction in the form

$$\frac{ax+b}{x^2-9}$$

Where a and b are integers. Work out the value of a and the value of b.

9) Show that

$$\frac{3x+6}{x^2-3x-10} \div \frac{x+5}{x^3-25x}$$

Simplifies to ax where a is an integer.